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DSC 680 – Applied Data Science

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Week 11

Toyota Used Car Market Analysis

Business Problem:

This project analyzes the Toyota used car market to identify the key factors that influence vehicle pricing and resale value. The analysis examines trends related to mileage, fuel type, transmission type, vehicle age, and model characteristics using several analytical and visualization methods in Python.

The goal of the project is to better understand how different vehicle characteristics impact the resale value of Toyota vehicles in the used car market. The project will also look at the pricing patterns and relationships between variables through statistical analysis and predictive modeling.

The used market uses a variety of factors to price their vehicles, including vehicle age, mileage, fuel type, transmission type, engine size, and model popularity. Understanding which factors have the greatest impact on pricing, purchasing, and inventory decisions.

This project will analyze the Toyota used car data to identify pricing trends and determine which variables have the most influence on resale value. The findings from the analysis may provide insights into how vehicle characteristics affect market prices and support more data-driven decision-making within the automotive industry.

Background / History:

The used car market represents a large portion of the automotive industry. The value of a vehicle declines over time due to depreciation, increased mileage, wear and tear, and consumer needs. However, each vehicle depreciates differently over time.

Toyota vehicles have historically maintained strong resale values because of their reputation for reliability, fuel efficiency, and durability. Since Toyota vehicles often retain their value, they are a good example to assess the current dataset to study pricing trends within the used vehicle market.

Statistical methods and machine learning techniques have become effective approaches for analyzing pricing behaviors and forecasting vehicle values. Businesses also use predictive analytics to improve management, establish pricing strategies, and identify market trends.

Data Explanation:

Dataset Source:

Kaggle Dataset: Toyota Used Car Market Insights

Dataset Link:

<https://www.kaggle.com/datasets/anassarfraz13/toyota-used-car-market-insights?resource=download>

Dataset File:

Toyota.csv

The dataset contains information regarding Toyota vehicles manufactured between 1998 and 2020, including 6,738 observations and 9 variables. The analysis will examine all years included in the dataset to explore pricing behavior and identify patterns associated with resale value.

Variables included in the dataset:

1. Price – Vehicle resale price
2. Model – Toyota vehicle model
3. Year – Manufacturing year
4. Transmission – Vehicle transmission type
5. Mileage – Distance driven
6. Fuel Type – Fuel category
7. Tax – Road Tax
8. MPG – Fuel efficiency
9. Engine Size – Vehicle engine size

Data Preparation:

The dataset was originally clean when it was first downloaded. Some preprocessing steps were still performed to check the data quality and consistency. The preprocessing steps included examining missing values, verifying variable formatting, evaluating summary statistics, and creating a new variable representing vehicle age.

Vehicle age was calculated using; $\text{Vehicle Age} = \text{Current Year} - \text{Manufacturing Year}$

The new variable allowed vehicle depreciation trends to be examined more effectively.

Methods:

Several analytical techniques were used during this study to examine the factors that influence Toyota vehicle resale prices.

Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) was performed to better understand the structure of the dataset and identify trends, distributions, and relationships between variables. Summary statistics and data inspection techniques were used to examine the variable and determine whether preprocessing was required.

Data Visualization:

Several visualizations were created using Python libraries including Matplotlib and Seaborn. Visualizations included histograms, scatter plots, box plots, and heatmaps to identify trends and relationships among variables.

Correlation Analysis:

Correlation analysis was performed to examine relationships between numerical variables and determine which factors had stronger associations with vehicle pricing.

Predictive Modeling:

A linear regression model was used to estimate Toyota vehicle prices based on selected variables, including mileage, vehicle age, fuel type, transmission type, engine size, and model type. Categorical variables were transformed into numerical values using one-hot encoding. The dataset was divided into training and testing datasets using an 80/20 split.

Analysis:

The visualizations and statistical analysis identified several important relationships within the Toyota used vehicle dataset.

The distribution of vehicle prices showed that most vehicles were concentrated within lower and moderate price ranges, while relatively few vehicles appeared at extremely high price levels.

The relationship between mileage and price showed a negative trend. As mileage increased, vehicle prices generally decreased, suggesting that higher vehicle usage contributes to lower resale values.

Vehicle age also demonstrated a negative relationship with resale price. Older vehicles were sold at lower prices compared to newer vehicles, supporting the expected depreciation value of older cars.

Differences in fuel type and vehicle characteristics also contributed to price variation among vehicles. Certain vehicle categories appeared to retain value more effectively than others.

Correlation analysis showed meaningful relationships among variables such as vehicle age, mileage, engine size, and price.

A linear regression model was developed to evaluate predictive performance using selected vehicle characteristics.

Model results:

Mean Absolute Error (MAE): 1,101.89

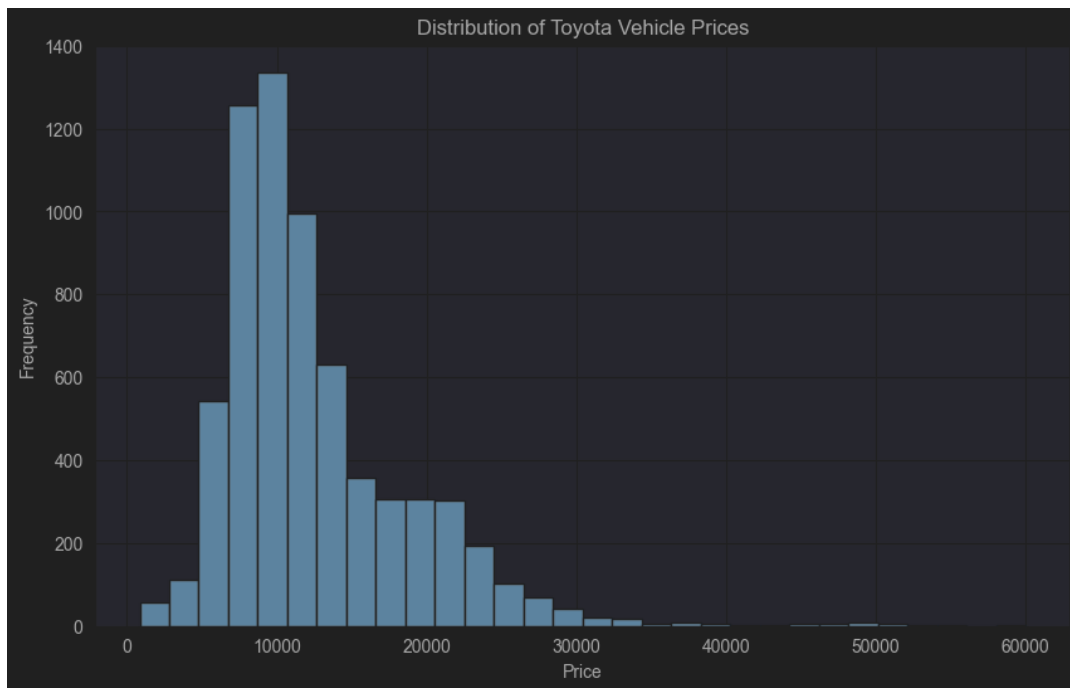
R² Score: 0.932

The Mean Absolute Error indicated that the model predictions differed from actual vehicle prices by approximately \$1,102 on average.

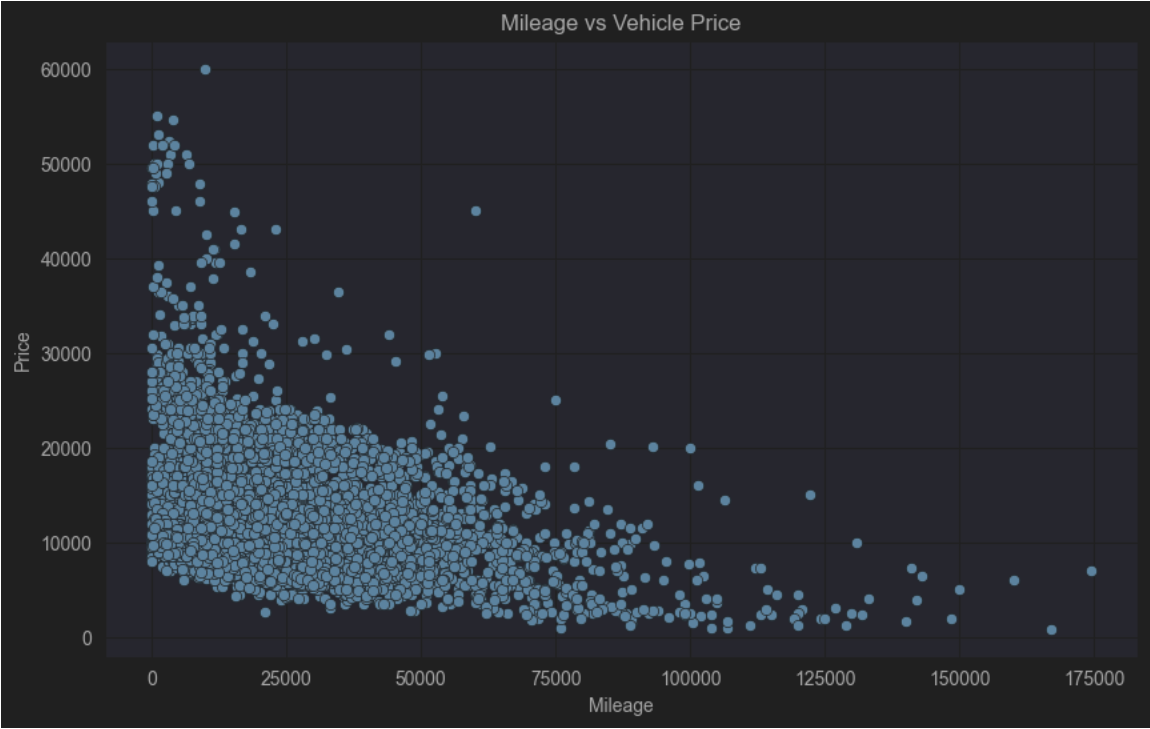
The R² score of 0.932 indicates that approximately 93.2% of the variation in vehicle prices was explained by the variables included in the model. Suggesting that there was strong predictive performance.

Visualizations from Python:

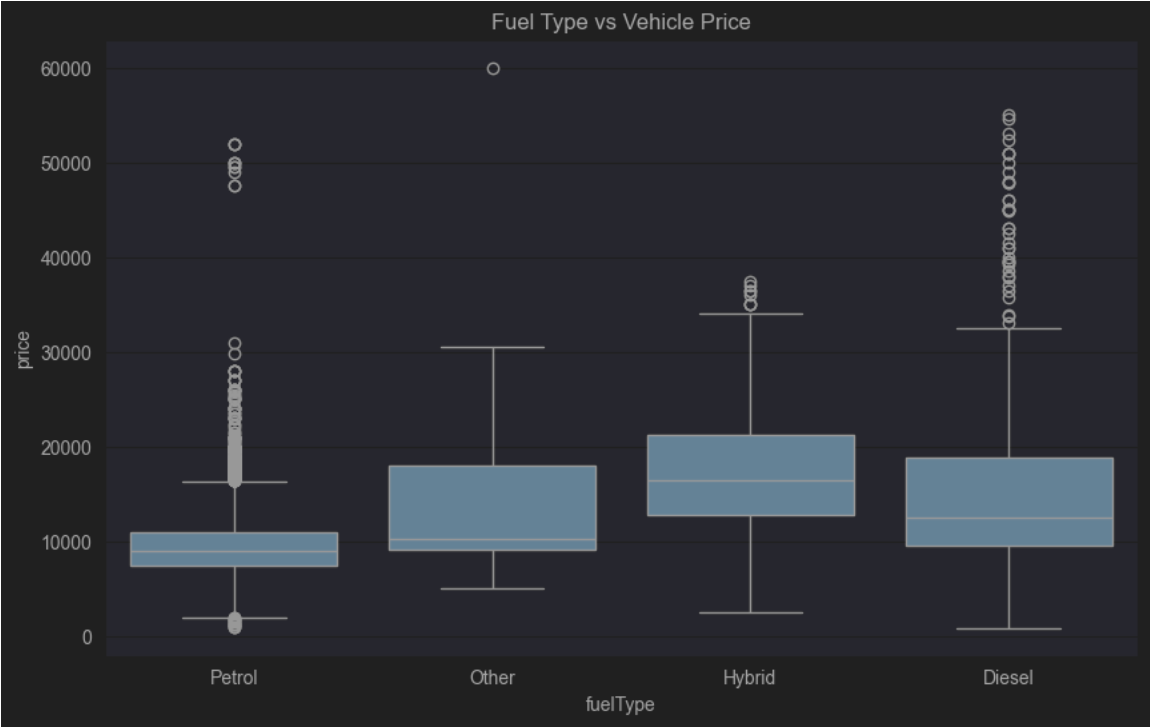
Visual 1: Distributions of Toyota Vehicle Prices



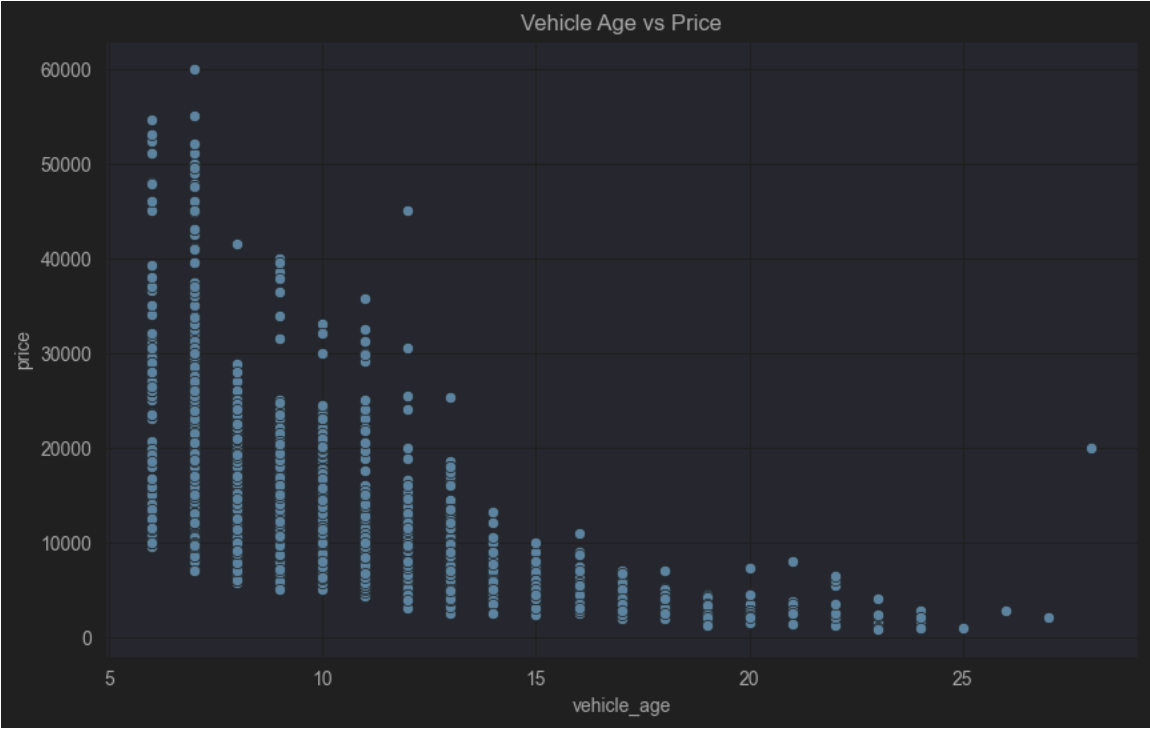
Visual 2: Mileage vs Vehicle Price Scatter Plot

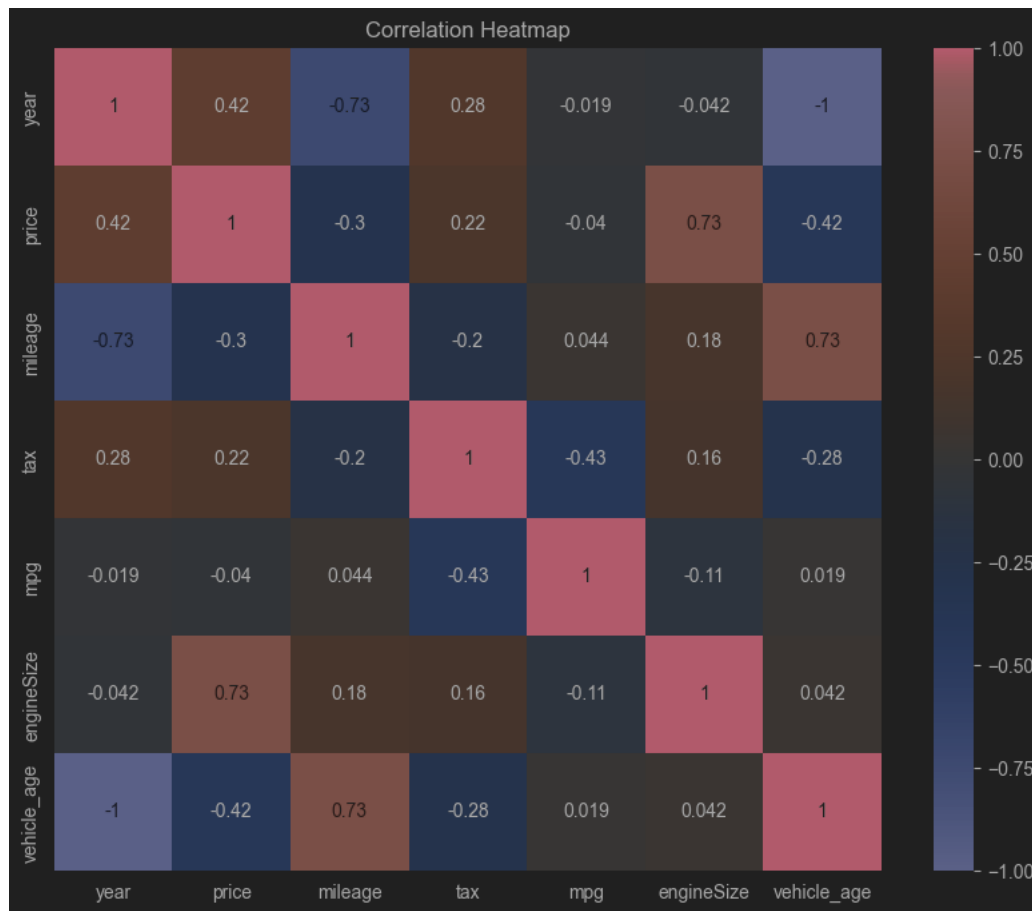


Visual 3: Fuel Type vs Vehicle Price Box Plot



Visual 4: Vehicle Age vs Vehicle Price Relationship





Conclusion:

The results of this study indicate that several vehicle characteristics significantly affect Toyota resale values. Variables including mileage, vehicle age, engine size, fuel type, transmission type, and model characteristics all contributed to pricing behavior within the used vehicle market.

The predictive model performed well with an R^2 value of 0.932, showing that the selective variables explained a large percentage of price variation. Along with the Mean Absolute Error of approximately \$1,102 suggests that predictive values remained relatively close to actual market prices.

The findings suggest that predictive analytics can provide valuable insights into vehicle pricing patterns and support data-driven decision-making within the automotive industry.

Assumptions:

Some of the assumptions that were made during the analysis were that vehicle information within the dataset was assumed to be accurate. The market conditions remained relatively stable throughout data collection. The selected variables adequately represented factors affecting resale value. Any missing variables didn't change the findings, found during the study.

Limitations:

Some of the limitations that should be considered during the study are that the dataset only includes Toyota vehicles. Some external factors, such as accident history, location, and economic conditions, were unavailable. Any reported data made by sellers may contain inaccuracies. Some regional market differences may not be represented.

Challenges:

Some challenges for this project could include handling missing or inconsistent data, identifying any outliers in vehicle pricing, and selecting the most appropriate variables for predictive analysis. Another limitation is that the analysis focuses only on Toyota vehicles and does not include comparisons with other automotive brands. The findings may therefore be specific to Toyota's used vehicle market rather than the broader used car industry.

Future Uses / Additional Applications:

This project has several potential future applications within the automotive industry. Future studies could expand the analysis to include different car manufacturers, instead of focusing only on Toyota vehicles. This project could also be used to develop dealership pricing recommendation systems to assist with establishing fair and competitive prices. Additional applications could include creating inventory management tools to help with dealership stock levels and forecasting future vehicle market trends.

Recommendations:

Based on the results of the analysis, several recommendations can be made. Dealerships should consider using predictive models to support pricing decisions and improve consistency when determining vehicle values. Consumers should also be looking at the car's mileage and vehicle age to estimate its value. Future studies should also include additional variables such as geographic location, accident history, and economic conditions since these factors can also have an impact of the vehicle's price.

Implementation Plan:

For the implementation plan, the goal is to continue collecting vehicle pricing information over time. Expand data collection to include additional manufacturers and test additional machine learning models for comparison. Also, evaluate model performance and prediction accuracy. Deploying predictive tools to support pricing and inventory decisions.

Ethical Consideration:

Potential ethical concerns include dataset bias, inaccuracies in seller-reported vehicle information, regional market differences, and changes in economic conditions over time. The dataset may not fully represent all Toyota vehicle sales or broader automotive market conditions.

Predictive modeling may unintentionally reinforce pricing assumptions or biases if important external market variables are excluded from the analysis. The findings should be interpreted carefully and shouldn't be used as the only factor when making pricing or purchasing decisions.

Questions:

1. Why was Toyota selected instead of multiple manufacturers?

Toyota vehicles are known for reliability and strong resale value, making them a good dataset for analyzing pricing behavior.

2. Which variable had the strongest influence on price?

Mileage and vehicle age showed the strongest negative relationships with price.

3. Why was linear regression selected?

Linear regression is interpretable and effective for continuous prediction problems. It works well when identifying relationships between variables.

4. How accurate was the prediction model?

The R^2 score indicates how much variation in vehicle price is explained by the model. A value of 0.932 means the model explained 93.2% of price variation. The Mean Absolute Error (MAE) was approximately \$1,102, meaning predictions were generally close to actual prices.

5. Did outliers affect the analysis?

Some high-priced vehicles existed, but overall trends remained consistent across the dataset. So, outliers didn't significantly impact the overall findings.

6. Why was vehicle age created as a new variable?

Vehicle age directly measures depreciation. So, it is easier to interpret in pricing analysis than the manufacturing year alone.

7. How does mileage influence resale value?

Mileage showed a negative relationship with resale value. As mileage increased, vehicle prices generally decreased because higher mileage often reflects more wear and tear than reduced vehicle lifespan.

8. Would adding location data improve predictions?

Yes, vehicle prices can change depending on where the vehicle is located. Due to regional and local market demand. Along with climate conditions, taxes, time of year, and consumer preferences. Location data could improve prediction accuracy.

9. Could this model be used for other vehicle brands?

Yes, used car dealers could apply this to their current stock of Toyota vehicles. If they used it for other vehicle brands, the model would need to be retained using those specific brands. Since each brand differs in pricing and depreciation patterns.

10. How could dealerships use these results?

Dealerships could use predictive analytics to support pricing decisions, improve inventory management, estimate trade in values, and identify vehicles that are more likely to retain value over time.

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